

**Project Design Phase-II**  
**Solution Requirements (Functional & Non-functional)**

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 28 June 2026                                 |
| Roll No       | 23A51A4229                                   |
| Project Name  | Metro Ticket Generating System in ServiceNow |
| Maximum Marks | 4 Marks                                      |

**Functional Requirements:**

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                       |
|--------|-------------------------------|----------------------------------------------------------------------------------------------------------|
| FR-1   | Metro Ticket Booking          | Select Source Station<br>Select Destination Station<br>Choose Journey Date<br>Enter Number of Passengers |
| FR-2   | Fare Calculation              | Calculate Fare Based on Route<br>Display Fare per Passenger<br>Calculate Total Fare Automatically        |
| FR-3   | Ticket Request Submission     | Submit Metro Ticket Request<br>Generate Request Number (REQ)<br>Generate Requested Item (RITM)           |
| FR-4   | Ticket Confirmation           | Display Booking Confirmation<br>Show Ticket Details<br>Provide Ticket Reference Number                   |
| FR-5   | Notification Management       | Send Booking Confirmation Email<br>Send Ticket Details to User<br>Notify Successful Booking              |
| FR-6   | Digital Ticket Generation     | Generate Digital Ticket<br>Create QR Code for Ticket<br>Attach QR Code to Ticket                         |

**Non-functional Requirements:**

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description                                                                                                                                                 |
|--------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1  | <b>Usability</b>           | The system should provide a simple, intuitive, and user-friendly interface that allows passengers to book metro tickets easily without technical knowledge. |
| NFR-2  | <b>Security</b>            | The system should ensure secure access through authenticated user accounts, role-based access controls, and secure data transmission using HTTPS protocols. |

|       |                     |                                                                                                                                                  |
|-------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-3 | <b>Reliability</b>  | The system should consistently generate accurate fare calculation confirmations with minimal errors or downtime.                                 |
| NFR-4 | <b>Performance</b>  | The system should calculate fares, process ticket bookings, and generate confirmations within a few seconds to provide a smooth user experience. |
| NFR-5 | <b>Availability</b> | The application should be accessible 24×7 through the service now availability to users.                                                         |
| NFR-6 | <b>Scalability</b>  | The system should support an increasing number of users, ticket bookings, and metro routes without significant degradation in performance.       |